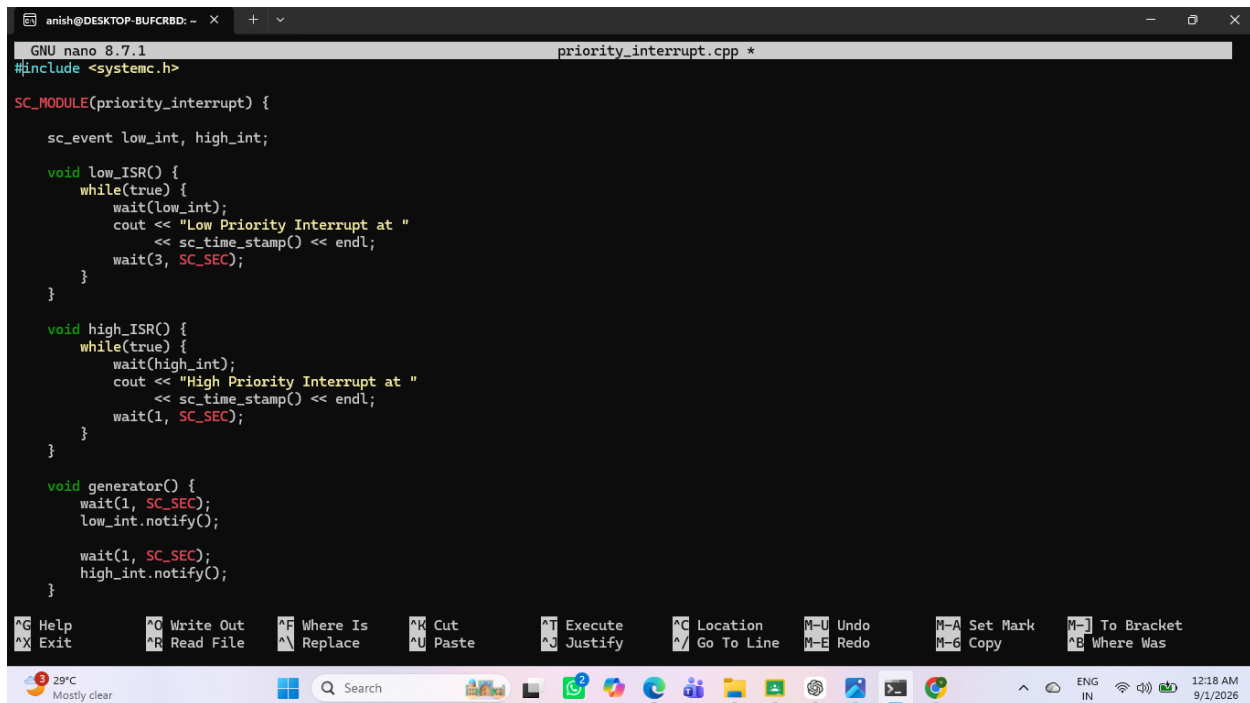


Exp No: 29

PRIORITY-BASED INTERRUPT HANDLING



```
GNU nano 8.7.1 priority_interrupt.cpp *
#include <systemc.h>

SC_MODULE(priority_interrupt) {

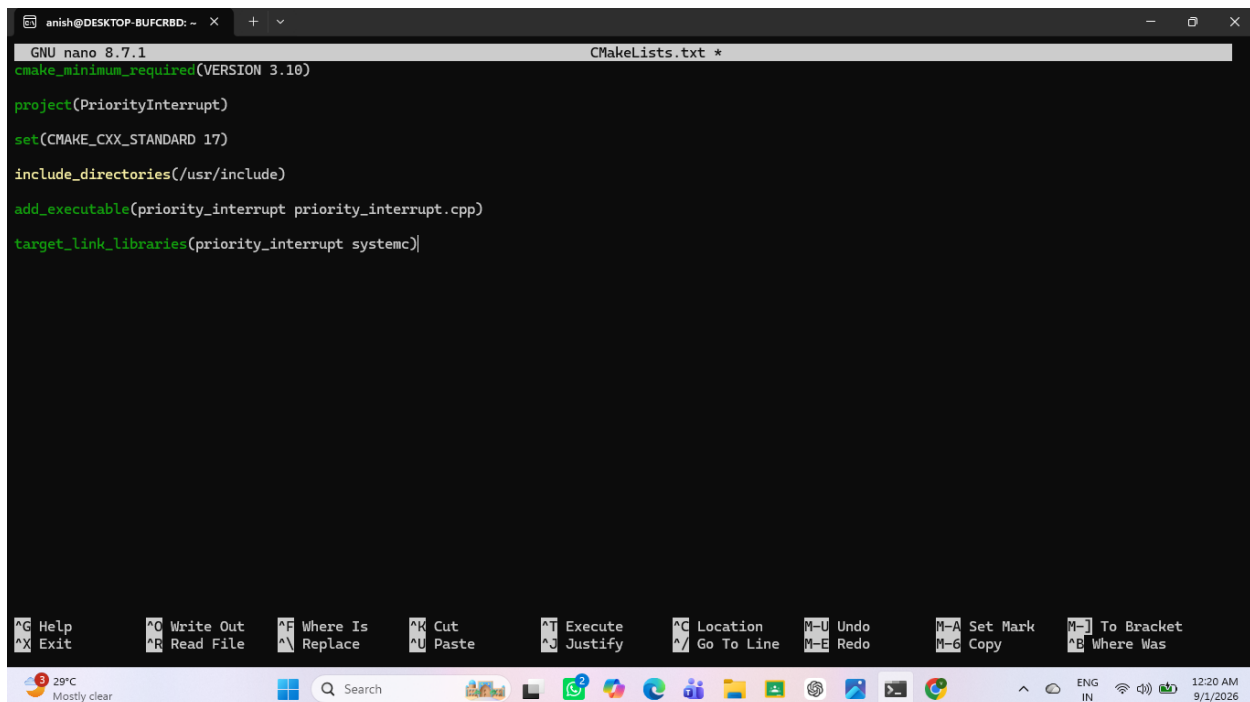
    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }

    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void generator() {
        wait(1, SC_SEC);
        low_int.notify();

        wait(1, SC_SEC);
        high_int.notify();
    }
}
```



```
GNU nano 8.7.1 CMakeLists.txt *
cmake_minimum_required(VERSION 3.10)

project(PriorityInterrupt)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(priority_interrupt priority_interrupt.cpp)

target_link_libraries(priority_interrupt systemc)
```

```
anish@DESKTOP-BUFCRBD: ~ - X + v
anish@DESKTOP-BUFCRBD:~/systemc_test$ ls priority_interrupt.cpp
priority_interrupt.cpp
anish@DESKTOP-BUFCRBD:~/systemc_test$ cd build
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ rm -rf CMakeCache.txt CMakeFiles Makefile cmake_install.cmake
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ cmake ..
-- The C compiler identification is GNU 15.2.0
-- The CXX compiler identification is GNU 15.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (1.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/anish/systemc_test/build
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ make
[ 50%] Building CXX object CMakeFiles/priority_interrupt.dir/priority_interrupt.cpp.o
[100%] Linking CXX executable priority_interrupt
[100%] Built target priority_interrupt
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ ./priority_interrupt

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ |
```